

Sifan Tao

Curriculum Vitae

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Education

- 2023 – Present **PhD Student in Statistics**, *University of Wisconsin-Madison*.
- 2019 – 2023 **BS in Statistics, with a second major in mathematics**, *University of Virginia*.
with distinction, Phi Beta Kappa
- Fall 2020 **UVA at Fudan: Go Local**, *School of Data Science, Fudan University*.

Publications

Journal Articles

- forthcoming **Sifan Tao** and Peter Chien. Group representative Latin hypercube design. *SIAM/ASA Journal of Uncertainty Quantification*, forthcoming.

Submitted

Qianxi Xing Ziqi Yang, **Sifan Tao** and Fang Liu. Spatiotemporal Origin and Dynamics of Sodium Plating in Sodium-Ion Batteries.

Sifan Tao and Peter Chien. Robust Latin Hypercube Design for Computer Experiments with Potential Failure Regions.

Xiaoyu Lei*, **Sifan Tao***, Peter Chien, and Joshua Cape. Overlapping sliced Latin hypercube designs for fusing multiple computer models. *: equal contribution.

Presentations

Conference Poster Presentations

- May 2025 **Uncertainty Quantification and Machine Learning for Complex Physical Systems**, *Variance Reduction under Grouped Variables in Sobol Models*, Institute for Mathematical and Statistical Innovation, Advised by Peter Chien.

Invited Talks

- November 2021 **Ohio State Sports Analytics Association Conference 2021**, *Modeling Stochastic Soccer Game Dynamics using Potential Functions*, Ohio State University, Joint work with Nicholas Ho, Adhvaith Vijay.
- October 2021 **Carnegie Mellon Sports Analytics Conference 2021**, *An Interpretable Method of Learning Stochastic Game Dynamics*, Carnegie Mellon University, Joint work with Nicholas Ho, Adhvaith Vijay.

Invited Guest Lectures

- October 2022 **STAT4800 Advanced Sport Analytics**, *Modeling Stochastic Soccer Game Dynamics using Potential Functions*, University of Virginia, Invited by Professor Jordan Rodu.
Undergraduate course on sport analytics.

Industry Experience

- Summer 2026 **Data Science PhD Intern**, *Capital One*, Upmarket Acquisition Behavioral, McLean, VA.

Student Research Experience

- Fall 2024 – **Graduate Research Assistant**, *Leverage-based Orthogonalizing EM Method*, Advised by Peter Chien.
Present University of Wisconsin-Madison
- Fall 2021 – **Undergraduate Research Assistant**, *Core-Periphery Trading Networks*, Advised by Chaojun Wang.
Fall 2022 The Wharton School, University of Pennsylvania
- Fall 2021 – **Undergraduate Research Course**, *Bayesian Propensity Scores, Sentiment analysis of Covid-19 tweet*, Advised by Xin Tong.
Spring 2022 University of Virginia
- Fall 2021 **Undergraduate Independent Study Course**, *Causal Inference*, Advised by Jordan Rodu.
University of Virginia
- Summer 2021 **Summer Undergraduate Research Experience**, *Modeling Stochastic Soccer game Dynamics using Potential functions*, Advised by Konstantinos Pelechrinis.
Carnegie Mellon Sports Analytics Camp 2021
- March 2021 – **Undergraduate Research Assistant**, *Why Trade Over-the-counter? When Investors Want Price Discrimination*, Advised by Chaojun Wang.
April 2021 The Wharton School, University of Pennsylvania

Service

Professional Service

Reviewer Journal of Statistical Planning and Inference

Teaching

Courses Served as Teaching Assistant at University of Wisconsin-Madison

- Fall 2026 **STAT424 Experimental Design.**
- Spring 2025 – **STAT340 Data Science Modeling II.**
Spring 2026
- Fall 2024 **STAT240 Data Science Modeling I.**
- Spring 2024 **STAT371 Introductory Applied Statistics for the Life Science.**
- Fall 2023 **STAT324 Introductory Applied Statistics for Engineers.**

Courses Served as Teaching Assistant at University of Virginia

- Fall 2021, **STAT3080 From Data to Knowledge.**
Fall 2022

Computer Skills

Proficient R

Intermediate Python

Beginner Shell, Java, SAS

Language

Bilingual Chinese, English

Beginner Japanese